

Deployment, Documentation & Final Presentation

1) Troubleshooting

Flow Execution & Logs:

Verified end-to-end execution of the Borrow Request Approval Flow using Flow Designer Execution Details. Checked trigger conditions, approvals, and book issuance updates.

← ↻ 🔒 https://dev280823.service-now.com/now/nav/ui/classic/params/target/u_borrow_request.do... 📄 ⭐ ⚙️ | ⭐ 👤 ⋮ 🗨 Chat ▾

servicenow All Favorites ⋮ Borrow Request - C... ☆ 🔍 🌐 🗨 📎 ? 🛎 👤

< ≡ Borrow Request New record 📎 ⚙️ ⋮ Submit

Requested By Alejandra Prenatt 🔍 ⓘ Book Created 2026-07-06 10:17:4 🔍 ⓘ

Request Date 2026-07-07 10:17:50 📅

Status Requested ▾

Submit ⓘ

Workflow Studio Borrow Request Approval F Flow • Global Borrow Request Approval F Flow execution • None + 👤

EXECUTION DETAILS Borrow Request Approval Flow 🔁 Test Run - Completed • Open flow Open context record

Show Action Details

Run as: System Administrator Open flow logs 📄

	State	Start time	
FLOW STATISTICS	Completed	2026-07-06 10:20:10	289ms

TRIGGER

🟢 Borrow Request Created or Updated Open current record ⓘ

ACTIONS

1 🟢 Ask For Approval Core Action Completed 2026-07-06 10:20:10 32ms

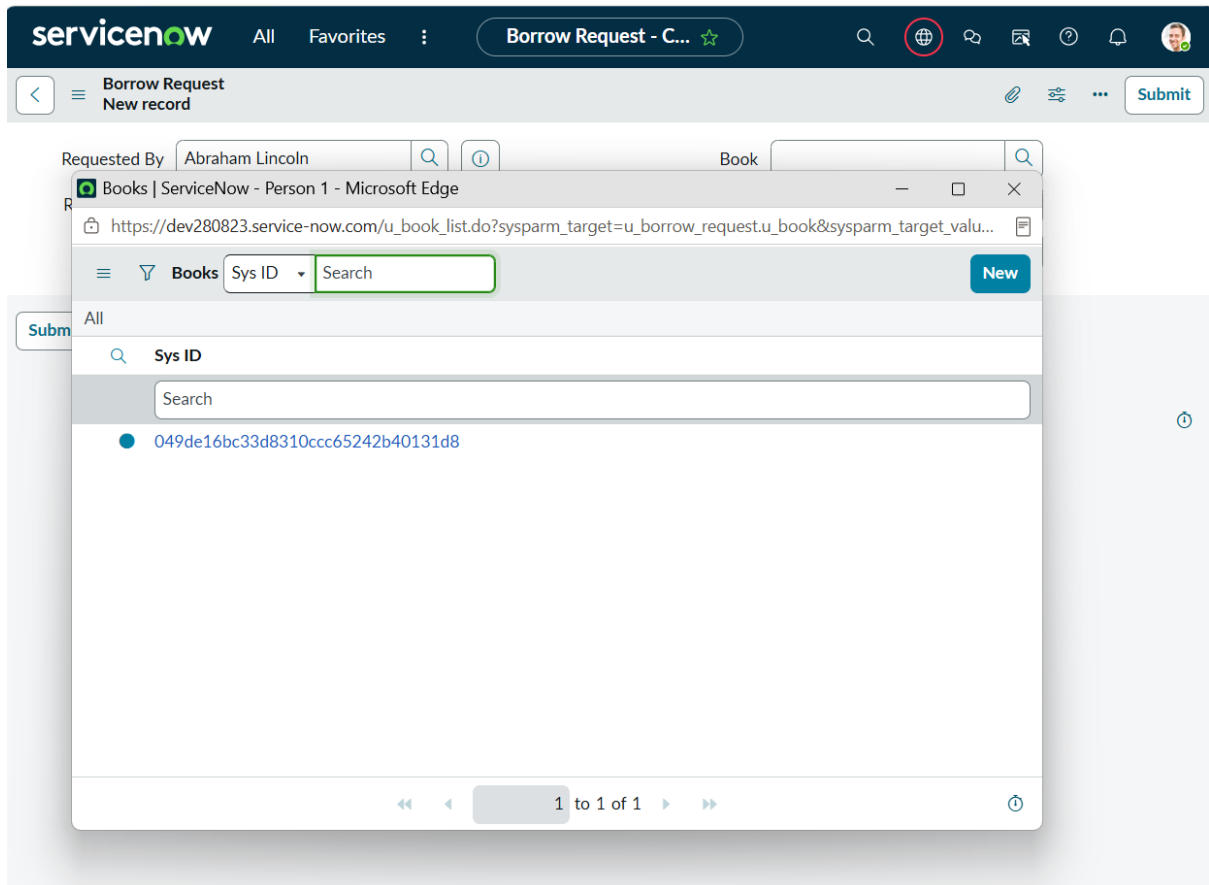
2 🟢 Update Record Core Action Completed 2026-07-06 10:20:10 5ms

3 🟢 Update Record Core Action Completed 2026-07-06 10:20:10 166ms

4 🟢 Send Email Completed 2026-07-06 10:20:11 86ms

ERROR HANDLER

Reference Qualifier Issue:

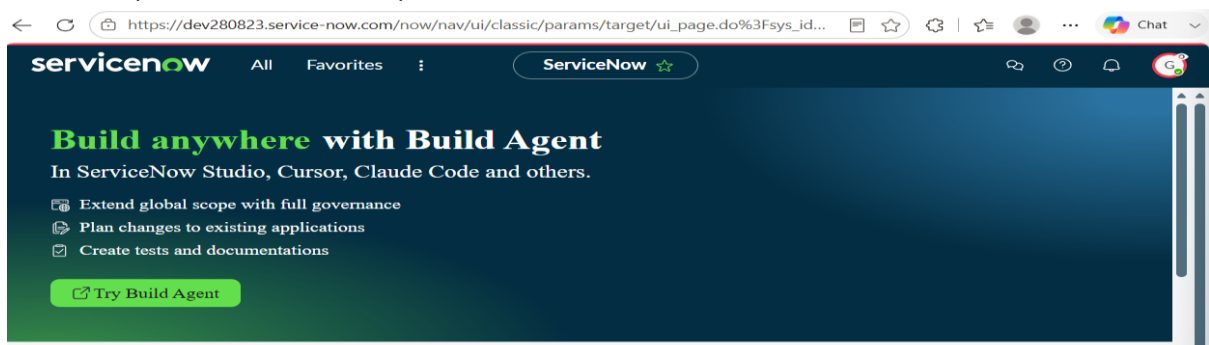


UI Policy Behavior

Tested Borrow Request form to ensure Return Date became mandatory when Status = Issued. Verified field visibility and mandatory settings across portal and platform views.

This screenshot shows the 'Borrow Request' form with the 'Status' field set to 'Approved'. The 'Requested By' field is populated with 'Abel Tuter'. The 'Request Date' is '2026-07-08 10:25:02'. The 'Return Date' field is marked with a red asterisk, indicating it is mandatory. The 'Submit' button is visible at the bottom.

ACL Test (Student Restricted)



← ↻ https://dev280823.service-now.com/u_book.do?sys_id=049de16bc33d8310ccc65242b40131... ☆ ⚙️ | ☆ 👤 ... Chat

< ≡ Book test2 ... ↑ ↓

Title test2

Author

ISBN

Status Available

ⓘ

< ≡ Book

Export >

Create Favorite

Copy URL

Reload form

ISBN

Status Available

Show/Hide Template Bar

Show/Hide Template Bar

ⓘ Show/Hide annotations

Tags

Add Tag

Report Accuracy

< ≡ Report Most Borrowed Books ⚙️ + ⚙️ ... Update View Report Edit Report Delete ↑ ↓

Please configure this report using Edit Report, and create new reports using View / Run.

Application Global ⓘ

Title Most Borrowed Books

Table Borrow Request [u_borrow_request] ▼

Field Name u_book

Type Bar

Chart size Large

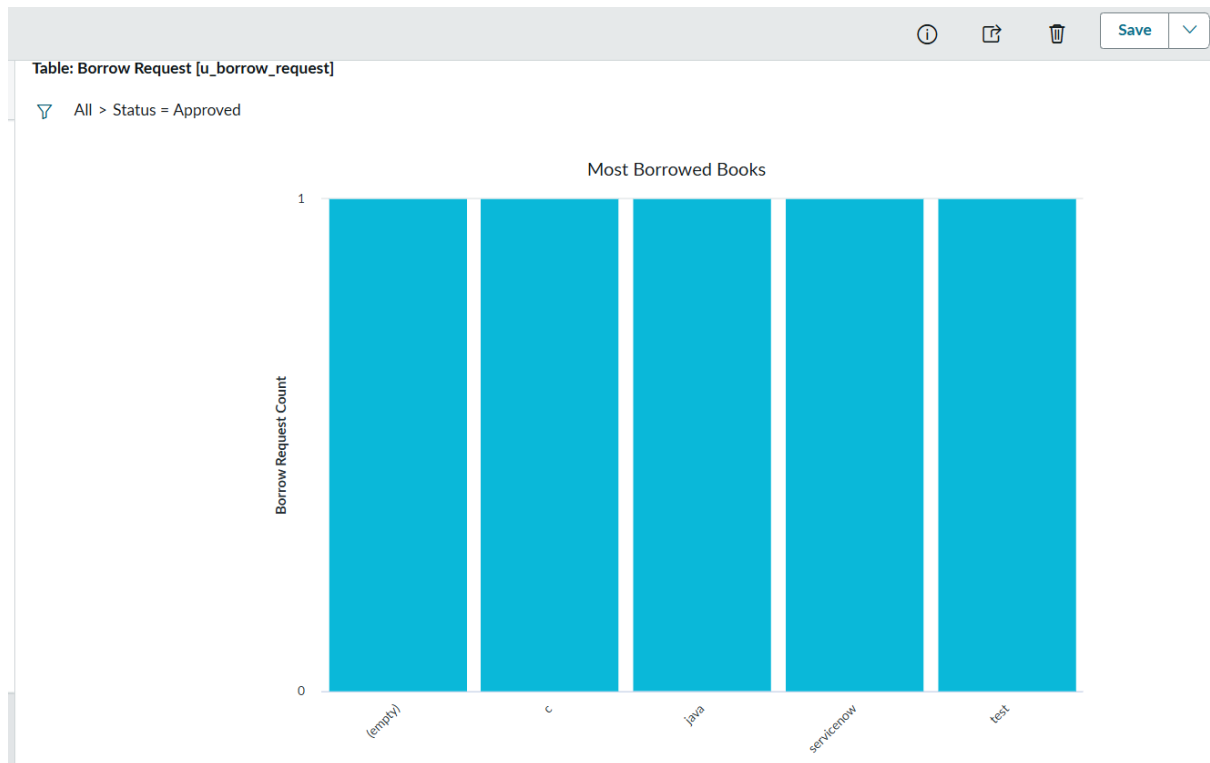
User 6816f79cc0a8016401c5a33be04be441

Filter Add Filter Condition Add OR Clause

Status ▼ is ▼ Approved ▼ AND

Roles

Update View Report Edit Report Delete



2) Adherence to Timelines

The Smart Library Request Workflow project was completed according to the planned development schedule. Each milestone was executed in the defined sequence, ensuring timely completion of all functional and technical requirements. The project activities included requirement analysis, application development, testing, security implementation, and documentation.

Milestones Completed

- Requirement analysis and project planning were completed.
- User roles (Student and Librarian) were created and configured.
- Book and Borrow Request tables were designed with the required fields.
- Reference Qualifier was implemented to display only available books.
- Borrow Request Approval Flow was developed using Flow Designer.
- UI Policies were configured to make the Return Date mandatory after approval.
- Access Control Lists (ACLs) were created for the Book and Borrow Request tables to provide role-based security.
- The "Most Borrowed Books" report was created and validated.
- Functional testing and troubleshooting were performed to verify workflows, UI policies, reports, and security.
- Final documentation and project screenshots were prepared.

Timeline Status

Phase	Status
Requirement Analysis & Planning	Completed
Backend Development & Configuration	Completed
Development	Completed
Data Migration, Testing & Security	Completed
Documentation	Completed

Phase	Status
Final Validation	Completed

Conclusion

All planned milestones were completed within the allocated timeline. Regular testing was performed after each development phase to identify and resolve issues early. The project was successfully delivered with all required

3) Innovation

The Smart Library Request Workflow introduces an automated and secure approach to library management by leveraging the capabilities of the ServiceNow platform. The project minimizes manual effort, improves operational efficiency, and enhances the overall user experience through workflow automation and role-based access control.

End-to-End Automation

The complete book borrowing process is automated using Flow Designer, enabling borrow requests to be submitted, approved, and processed without manual intervention. Once a request is approved, the system automatically updates the book status to Issued, updates the borrow request status, and sends email notifications to the student.

Optimized Request Handling

A Reference Qualifier is implemented on the Book reference field to display only books with the Available status. This prevents students from selecting books that have already been issued, eliminating duplicate borrow requests and improving data accuracy.

Enhanced Security

The application uses Access Control Lists (ACLs) to enforce role-based security. Students are permitted to create and view borrow requests, while librarians have permissions to approve requests, update book records, and manage library operations. This ensures secure access to sensitive data and prevents unauthorized modifications.

Improved User Experience

Dynamic UI Policies are implemented to simplify data entry. The Return Date field becomes mandatory only after a borrow request is approved, ensuring that users provide the required information only when necessary. This keeps forms clean, reduces unnecessary fields, and improves usability.

Reporting and Analytics

A "Most Borrowed Books" report is created to provide librarians with insights into borrowing trends. The report helps identify popular books, supports inventory planning, and enables data-driven decision-making for future library management.

Scalable Solution

The solution is built using standard ServiceNow features such as Custom Tables, Flow Designer, Reference Qualifiers, UI Policies, Access Control Lists, and Reports. This modular design makes the application easy to maintain, extend, and adapt for additional library features in the future.

Conclusion

The project demonstrates an innovative use of ServiceNow to build a modern library management solution that combines workflow automation, intelligent validation, secure access control, and analytical reporting. These enhancements improve operational efficiency, reduce manual work, and provide a reliable and user-friendly experience for both students and librarians.

4) Document Technical Blueprint

Variables

The Smart Library Request Workflow uses the following variables to capture and process borrowing information:

- **Book Title** – Reference field used to select the book to be borrowed.

- **Author** – Displays the author of the selected book.
- **Requested By** – Automatically populated with the currently logged-in user.
- **Borrow Date** – Automatically set to the current system date when the request is created.
- **Return Date** – Becomes mandatory only after the borrow request is approved and the book is issued.
- **Status** – Indicates the current state of the request (Requested, Approved, Rejected, Issued, Returned).
- **Comments** – Stores remarks entered by the librarian during approval or rejection.

Approval Use Cases

- When a student submits a borrow request, the system automatically initiates the **Borrow Request Approval Flow**.
- The request is assigned to the **Librarian** for approval.
- The librarian can either **Approve** or **Reject** the request.
- If the request is approved, the book status is updated to **Issued**, the borrow request status changes to **Approved**, and an email notification is sent to the student.
- If the request is rejected, the borrow request status is updated to **Rejected**, and the student receives a rejection notification.

Fulfillment Task Use Cases

- After approval, the librarian issues the requested book to the student.
- The system automatically updates the **Book** record status from **Available** to **Issued**.
- The **Borrow Request** record is updated with the approval status and issue details.
- The **Return Date** becomes mandatory through a UI Policy before the request is finalized.
- When the student returns the book, the librarian updates the book status to **Available** and the borrow request status to **Returned**.
- The workflow ensures that only available books can be requested and prevents duplicate borrow requests through the Reference Qualifier.

Technical Components Used

- **Custom Tables:** Book, Borrow Request
- **Reference Field:** Book reference in Borrow Request
- **Reference Qualifier:** Displays only books with **Available** status
- **Flow Designer:** Automates the borrow request approval process
- **UI Policy:** Makes the Return Date mandatory after approval
- **Access Control Lists (ACLs):** Implements role-based security for Students and Librarians
- **Reports:** Generates the **Most Borrowed Books** report for analytics

Expected Outcome

The Smart Library Request Workflow provides an automated, secure, and efficient library management process. It minimizes manual work by automating approvals, restricting access based on user roles, validating user input through UI Policies, and providing analytical reports to help librarians monitor borrowing trends. The modular design also allows the application to be easily extended with additional features in the future.

5)Document Setup Manual

The following setup procedure was followed to implement the **Smart Library Request Workflow** application in a new ServiceNow Personal Developer Instance (PDI).

1. Role Creation

- Created two user roles: **Student** and **Librarian**.
- Assigned appropriate permissions to each role.
- Created test users and assigned the required roles for application testing.

2. Table Creation

Created the following custom tables:

- **Book (u_book)** – Stores information about books available in the library.
- **Borrow Request (u_borrow_request)** – Stores book borrowing requests submitted by students.

3. Field Configuration

Configured the required fields for each table.

Book Table

- Title
- Author
- ISBN
- Status (Available, Issued)
- Description

Borrow Request Table

- Requested By
- Book (Reference to Book table)
- Borrow Date
- Return Date
- Status (Requested, Approved, Rejected, Returned)
- Comments

4. Reference Qualifier Configuration

Configured the **Book** reference field in the Borrow Request table to display only books with the **Available** status. This prevents students from requesting books that have already been issued.

5. Flow Designer Configuration

Created the **Borrow Request Approval Flow** in Flow Designer.

The flow performs the following actions:

- Triggers when a borrow request is created.
- Sends the request to the librarian for approval.
- Updates the Book status to **Issued** after approval.
- Updates the Borrow Request status to **Approved**.
- Sends an email notification to the student.
- Supports rejection handling when the request is not approved.

6. UI Policy Configuration

Created a UI Policy named "**Make Return Date Mandatory when Issued**".

The policy:

- Executes when the Borrow Request status becomes **Approved**.
- Makes the **Return Date** field visible and mandatory.
- Improves data validation by ensuring every issued book has a return date.

7. Access Control Configuration

Configured Access Control Lists (ACLs) to implement role-based security.

Book Table

- Read → Student, Librarian
- Write → Librarian
- Delete → Librarian

Borrow Request Table

- Create → Student
- Read → Student, Librarian
- Write → Librarian

- Delete → Librarian

8. Report Configuration

Created a report named "**Most Borrowed Books**".

The report:

- Uses the Borrow Request table as the data source.
- Groups records by Book.
- Counts approved borrow requests.
- Displays the results using a Bar Chart.

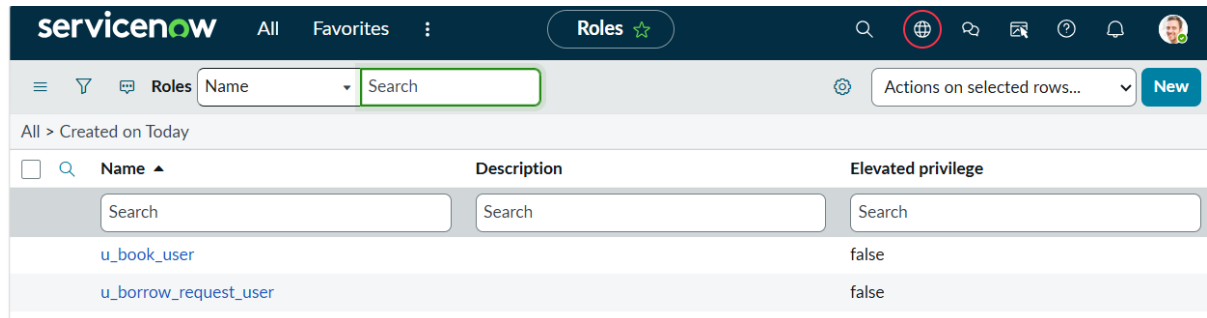
9. Testing and Validation

Performed functional testing to verify:

- Borrow request creation.
- Reference Qualifier functionality.
- Flow execution and approval process.
- UI Policy behavior.
- ACL restrictions for Student and Librarian roles.
- Report accuracy.

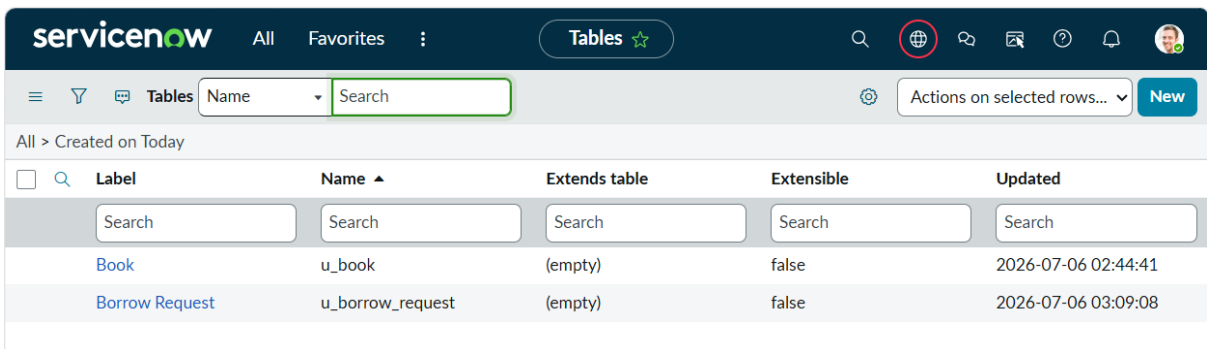
10. Documentation

Role creation



Name	Description	Elevated privilege
u_book_user		false
u_borrow_request_user		false

Table creation



Label	Name	Extends table	Extensible	Updated
Book	u_book	(empty)	false	2026-07-06 02:44:41
Borrow Request	u_borrow_request	(empty)	false	2026-07-06 03:09:08

Field Configuration

Table Borrow Request

DeleteUpdateDelete All Records

ColumnsControlsApplication Access

Table Columnsfor textSearch1 to 11 of 11New

Dictionary Entries

Column label	Type	Reference	Max length	Default value	Display
Created	Date/Time	(empty)	40		false
Sys ID	Sys ID (GUID)	(empty)	32		false
Updates	Integer	(empty)	40		false
Request Date	Date/Time	(empty)	100		false
Requested By	Reference	User	32		false
Updated by	String	(empty)	40		false
Updated	Date/Time	(empty)	40		false
Return Date	Date/Time	(empty)	100		false
Book	Reference	Book	32		false
Created by	String	(empty)	40		false
Status	Choice	(empty)	100	Requested	false
Insert a new row...					

Reference Qualifier

Dictionary Entry Book

Delete ColumnUpdate

Column labelBook

Column nameu_book

Function fieldRead onlyMandatoryDisplay

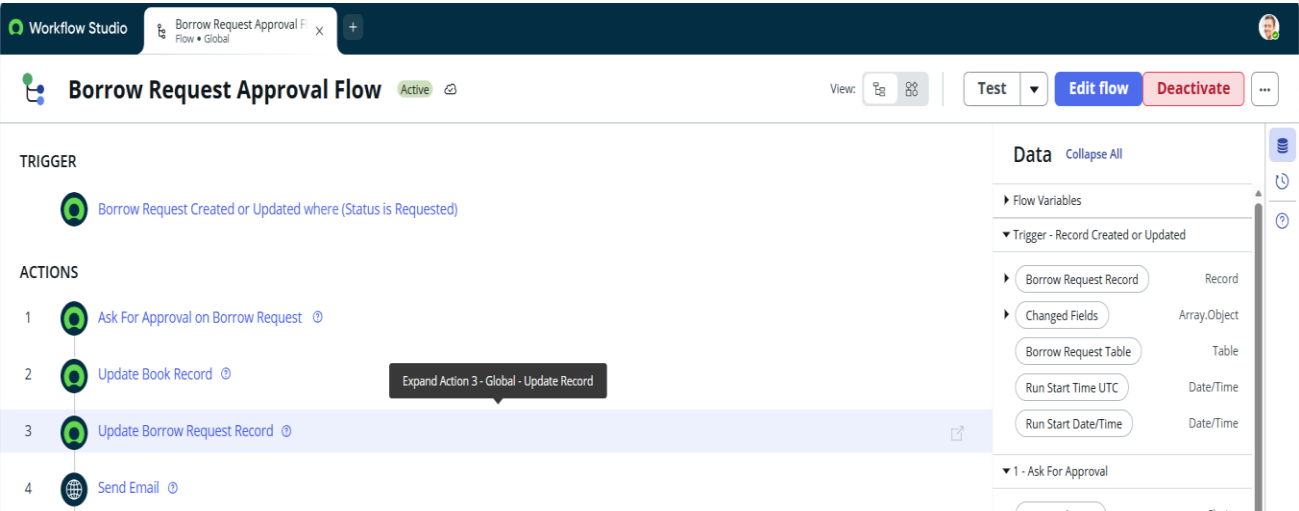
Reference SpecificationChoice List SpecificationDefault Value

The Reference field specifies what table this field displays values from.

ReferenceBook

Reference qual conditionAdd Filter ConditionAdd OR ClauseStatus is AvailableAN

Flow Designer



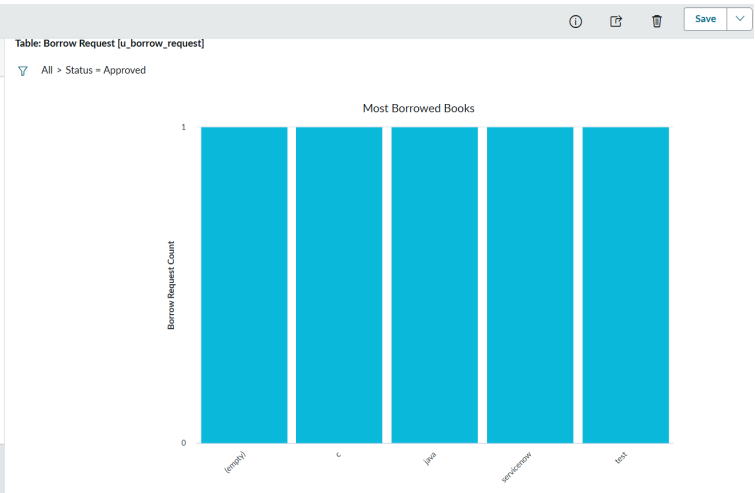
UI Policies

servicenow						
All Favorites UI Policies ☆						
UI Policies Order Search						
All > Created on Today						
Short description	Table	Conditions	Reverse if false	On load	Updated	Order
Search	Search	Search	Search	Search	Search	Search
Make Return Date mandatory when Issued.	Borrow Request [u_borrow_request]	u_status=300^EQ	true	true	2026-07-06 06:23:41	100

ACL configuration

Access Controls Name Search						
All > Created on Today						
Name	Decision Type	Operation	Type	Active	Updated by	Updated
Search	Search	Search	Search	Search	Search	Search
u_book	Allow If	create	record	true	admin	2026-07-06 02:44:42
u_book	Allow If	write	record	true	admin	2026-07-06 02:44:42
u_book	Allow If	delete	record	true	admin	2026-07-06 08:12:22
u_book	Allow If	write	record	true	admin	2026-07-06 08:11:19
u_book	Allow If	delete	record	true	admin	2026-07-06 02:44:42
u_book	Allow If	read	record	true	admin	2026-07-06 02:44:42
u_book	Allow If	read	record	true	admin	2026-07-06 08:10:11
u_borrow_request	Allow If	write	record	true	admin	2026-07-06 08:47:10
u_borrow_request	Allow If	read	record	true	admin	2026-07-06 02:45:29
u_borrow_request	Allow If	delete	record	true	admin	2026-07-06 02:45:29
u_borrow_request	Allow If	create	record	true	admin	2026-07-06 08:43:13
u_borrow_request	Allow If	delete	record	true	admin	2026-07-06 08:54:09
u_borrow_request	Allow If	read	record	true	admin	2026-07-06 08:44:39
u_borrow_request	Allow If	write	record	true	admin	2026-07-06 02:45:29
u_borrow_request	Allow If	create	record	true	admin	2026-07-06 02:45:29

Reports



Testing results

Workflow Studio Borrow Request Approval Flow			
EXECUTION DETAILS			
Borrow Request Approval Flow			
Show Action Details			
FLOW STATISTICS			
Run as: System Administrator			
Open flow logs			
TRIGGER			
Borrow Request Created or Updated			
ACTIONS			
1	Ask For Approval	Core Action	Completed
2	Update Record	Core Action	Completed
3	Update Record	Core Action	Completed
4	Send Email		Completed
ERROR HANDLER			

7)Visual Demonstration

The visual demonstration of the Smart Library Request Workflow has been comprehensively covered through the project documentation and the project demonstration video. All major implementation steps, configurations, and workflow executions are supported with screenshots and detailed explanations in the documentation, while the complete end-to-end functionality is demonstrated in the video.

The documentation includes visual evidence of:

- Role creation and user configuration.
- Book and Borrow Request table creation.
- Field configuration and Reference Qualifier implementation.
- Flow Designer configuration and execution.
- UI Policy configuration and testing.
- Access Control List (ACL) implementation.
- Report creation and generated output.
- Testing, troubleshooting, and validation results.

The project demonstration video showcases the complete workflow, including:

- Student borrow request submission.
- Librarian approval process.
- Automatic workflow execution.
- Book status updates.
- UI Policy behavior.
- Report generation.
- Role-based access control and security validation.

Since all screenshots and workflow demonstrations are already included in the documentation and demo video, they are not duplicated in this section to maintain clarity and avoid redundancy. The provided documentation and video together serve as the complete visual reference for the implementation and execution of the Smart Library Request Workflow.

8)Explanation Clarity

Documentation Structure

The Smart Library Request Workflow documentation is organized in a clear and logical sequence, making it easy to understand and reproduce the project. Each module is explained with its objective, implementation steps, screenshots, and expected outcomes.

The documentation follows the flow below:

- Requirement Analysis and Planning
- Backend Development and Configuration
- Development using Flow Designer
- Data Migration, Testing, and Security
- Deployment and Documentation
- Final Demonstration and Project Conclusion

Configuration Details

Each ServiceNow configuration is explained with clear headings and implementation details, including:

- Creation of Student and Librarian roles
- Creation of Book and Borrow Request tables
- Field configuration for both tables
- Reference Qualifier implementation for displaying only available books
- Borrow Request Approval Flow using Flow Designer
- UI Policy configuration for making the Return Date mandatory

- Access Control Lists (ACLs) for role-based security
- Report creation for "Most Borrowed Books"

Each configuration includes screenshots and a brief explanation of its purpose and functionality.

Workflow Explanation

The project workflow is presented in a step-by-step manner:

1. Student submits a borrow request.
2. The system validates that the selected book is available.
3. The request is automatically sent to the Librarian for approval.
4. The Librarian approves or rejects the request.
5. If approved, the book status changes to **Issued** and the borrow request status is updated.
6. The Return Date becomes mandatory through the UI Policy.
7. When the book is returned, the status is updated to **Returned** and the book becomes available again.

Realism & Quality

The Smart Library Request Workflow is designed to simulate real-world library operations using ServiceNow automation.

The solution includes:

- Role-based access for Students and Librarians.
- Automated approval workflow using Flow Designer.
- Validation to ensure only available books can be borrowed.
- UI Policies to enforce mandatory fields based on request status.
- Access Control Lists (ACLs) to secure application data.
- Reporting and analytics to identify frequently borrowed books.

The application was tested using multiple borrow requests, approval scenarios, and book return processes to ensure reliable performance and accurate workflow execution.

Scalability & Future Enhancements

The application is designed using standard ServiceNow features, making it easy to maintain and extend.

Future enhancements may include:

- Automatic email reminders for overdue books.
- SMS or mobile notifications for due dates and approvals.
- Online book reservation and waiting list functionality.
- Integration with barcode or QR code scanning for book issue and return.
- Dashboard analytics for borrowing trends, student activity, and overdue books.
- Integration with external digital library systems and academic resource platforms.
- Additional modules for laboratory equipment requests, study room booking, and digital resource management.

Conclusion

The Smart Library Request Workflow provides a secure, automated, and user-friendly solution for managing library borrowing activities. The documentation is presented in a structured manner with clear explanations, screenshots, and implementation details, enabling users to understand, deploy, and replicate the solution efficiently in any ServiceNow Personal Developer Instance (PDI).